

Simulation-Reality Deviation Module (SDMM) defines the governance

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Governed Space: Simulation-Reality Deviation

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Governed Space

Simulation-Reality Deviation

Canonical Definition

Simulation-Reality Deviation Module (SDMM) defines the governance conditions under which material differences between simulated behavior, conditions, states, events, relationships, trajectories, and outputs and their corresponding Operational Reality observations are identified, measured, localized, classified, contextualized, traced, and documented.

SDMM establishes the governed deviation layer between:

Simulation-Reality Comparison

and:

Reality Alignment.

Its purpose is not to require simulation and reality to be identical.

Its purpose is to determine:

where they differ, by how much, under which conditions, whether the deviation is systematic or conditional, and which part of the simulation-to-reality relationship is affected.

Operational Role

SDMM is the third internal governance space of the Simulation-to-Reality Correlation Standard (SRCS).

The progression is:

SRCM — Simulation-Reality Comparison Module

→ establishes comparable simulation and reality observations.

SCCM — Simulation Correlation Criteria Module

→ establishes the criteria against which correspondence is assessed.

SDMM — Simulation-Reality Deviation Module

→ identifies and structures material differences revealed by that comparison.

SRAM — Reality Alignment Module

→ governs how the simulation-reality relationship is reassessed or aligned in response to identified differences.

SCUM — Correlation Uncertainty Module

→ governs uncertainty remaining in the correlation relationship.

SDMM therefore asks:

Where does the simulation depart from reality, how material is that departure, and what does the deviation actually affect?

Module Operational Space

SDMM governs: simulation-reality deviation identification, deviation measurement, deviation localization, deviation classification, deviation magnitude, deviation direction, deviation persistence, deviation frequency, deviation distribution, temporal deviation, spatial deviation, behavioral deviation, state deviation, event deviation, trajectory deviation, response deviation, interaction deviation, environmental deviation, output deviation, threshold deviation, systematic deviation, conditional deviation, intermittent deviation, localized deviation, cumulative deviation, correlated deviation, deviation clustering, deviation dependency, deviation propagation, deviation attribution, unresolved deviation, deviation materiality, deviation

Module Function

SDMM prevents a simulation from being described as simply:

“close to reality”

or:

“different from reality”

without identifying the structure of that difference.

A useful simulation-reality assessment must be capable of distinguishing, where relevant:

what matched,

what deviated,

where the deviation occurred,

under which conditions it occurred,

how large it was,

whether it repeated,

and:

what downstream interpretation may be affected.

Deviation Object

Every material deviation should be capable of being represented as a governable object.

Conceptually:

Deviation ID

↓

Simulation Observation

↓

Reality Observation

↓

Correlation Criterion

↓

Observed Difference

↓

Deviation Classification

↓

Materiality

↓

Affected Simulation Element

↓

Known / Suspected Basis

↓

Uncertainty

↓

Governance Status

This prevents significant deviations from disappearing inside aggregate correlation metrics.

Deviation Identification

A Simulation-Reality Deviation exists when a materially comparable simulated and real-world observation differs according to an applicable correlation criterion.

The difference may concern: magnitude, direction, timing, sequence, frequency, duration, state, response, interaction, distribution, threshold crossing, failure mode, recovery behavior, or another materially relevant property.

Deviation ≠ Failure

SIMULOS® establishes:

A Simulation-Reality Deviation is not automatically a simulation failure.

A deviation may arise because: the simulation is simplified, reality contains unmodeled variables, measurements are imperfect, the Operational Reality state changed, boundary conditions differ, assumptions apply only conditionally, stochastic behavior exists, the comparison basis is incomplete, or the simulation genuinely misrepresents reality.

SDMM preserves the deviation before determining its consequence.

Deviation Magnitude

Where measurable, deviation magnitude should establish the size of the observed difference.

Magnitude may be expressed through: absolute difference, relative difference, percentage difference, distributional distance, timing offset, positional error, trajectory error, frequency difference, state mismatch, categorical difference, or another appropriate measure.

SIMULOS® does not prescribe one universal deviation metric.

Deviation Direction

Where relevant, SDMM should distinguish whether the simulation:

Overestimates Reality

Underestimates Reality

Leads Reality

Lags Reality

Diverges Bidirectionally

or exhibits another defined directional relationship.

Direction may be operationally more important than aggregate error magnitude.

Deviation Localization

A deviation should be localized as precisely as the available evidence permits.

Localization may identify: Simulation Object component, subsystem, agent, environment region, scenario, parameter range, operating condition, time interval, state transition, interaction, input region, or output dimension.

Global ≠ Local Correlation

A simulation may correlate strongly with reality overall while failing badly in a specific operational region.

Therefore:

Strong aggregate correlation must not erase material localized deviation.

Temporal Deviation

Temporal deviation exists where simulated behavior differs from reality in: initiation time, response latency, event ordering, duration, synchronization,

recovery time, transition speed, or temporal pattern.

A simulation may predict the correct event but at the wrong time.

That difference may be material.

Spatial Deviation

Spatial deviation concerns differences in: position, trajectory, distribution, physical location, movement, coverage, geometry, or spatial interaction.

This may be especially important in: robotics, autonomous vehicles, aerospace, orbital systems, manufacturing, logistics, and physical infrastructure simulation.

Behavioral Deviation

Behavioral deviation exists where the simulated system responds differently from the corresponding real system under materially comparable conditions.

Examples include: different action selection, different recovery behavior, different escalation behavior, different human response, different agent coordination, or different autonomous decision pathways.

Environmental Deviation

A simulation may reproduce the target system accurately while inadequately reproducing its environment.

SDMM therefore distinguishes:

System Deviation

from:

Environment Deviation.

Environmental deviation may concern: weather, terrain, friction, radiation, communication conditions, traffic, human behavior, physical noise, infrastructure state, or another environmental variable.

Interaction Deviation

Complex systems may behave correctly in isolation but diverge when interacting with: humans, other agents, infrastructure, external systems, physical environments, or unexpected events.

SDMM therefore governs deviations in interaction dynamics separately where material.

Trajectory Deviation

For systems whose behavior evolves over time, a small initial deviation may produce substantial downstream divergence.

Conceptually:

Initial Difference

↓

State Divergence

↓

Behavioral Divergence

↓

Trajectory Divergence

↓

Material Outcome Difference

SDMM should preserve this progression where observable.

Systematic Deviation

A Systematic Deviation repeatedly occurs in the same direction or under comparable conditions.

Examples:

simulation consistently underestimates braking distance

or:

simulation consistently overestimates communication reliability.

Systematic deviations may indicate a persistent representational or model relationship requiring further governance.

Conditional Deviation

A simulation may correlate well with reality under normal conditions but diverge under a specific condition.

For example:

Normal Temperature → Strong Correlation

Extreme Temperature → Material Deviation

The correct conclusion is not necessarily:

Simulation Poor

but potentially:

Correlation is conditionally bounded.

Intermittent Deviation

Some deviations occur irregularly.

This may arise from: stochastic processes, rare interactions, timing, emergent behavior, nondeterministic systems, or incomplete observation.

Intermittent deviations should not automatically be discarded as noise where their consequences may be material.

Deviation Frequency

Where relevant, SDMM may govern:

how often the deviation occurs.

A small deviation occurring continuously may be more important than a larger deviation occurring once.

Conversely, a rare deviation may still be critical where its consequence is catastrophic.

Deviation Materiality

Deviation magnitude alone does not determine materiality.

Conceptually:

Deviation Materiality

may depend upon:

Magnitude + Context + Frequency + Consequence + Scenario + Reliance Purpose.

A two-percent difference may be immaterial in one simulation and mission-critical in another.

Materiality Classification

Possible implementation states may include:

Negligible

Minor

Material

Major

Critical

or:

Materiality Undetermined.

These classifications are implementation-dependent rather than universal SIMULOS® thresholds.

Threshold Deviation

Where correlation criteria establish acceptable thresholds, SDMM should distinguish:

Within Correlation Boundary

from:

Outside Correlation Boundary.

However, passing a threshold does not prove universal simulation-to-reality equivalence.

Deviation Clustering

Individual deviations may reveal little when viewed separately.

SDMM may therefore identify:

Deviation Clusters

where multiple differences occur around: one subsystem, one scenario family, one operating condition, one environment state, one parameter region, or one behavioral pattern.

Clusters can expose simulation weaknesses hidden by aggregate metrics.

Deviation Propagation

A deviation in one simulated element may propagate through the simulation.

For example:

Sensor Timing Deviation

↓

State Estimation Deviation

↓

Decision Deviation

↓

Trajectory Deviation



Outcome Deviation

SDMM should preserve known propagation relationships rather than treating each downstream difference as independent.

Deviation Dependency

Some deviations exist only when multiple conditions combine.

For example:

Low Visibility Sensor Noise High Speed



Material Simulation-Reality Deviation

Each factor independently may show acceptable correlation.

The combination may not.

Deviation Attribution

Where sufficient evidence exists, a deviation may be associated with a probable basis such as: Reality Representation, assumption, model structure, environment configuration, parameterization, input data, scenario design, execution condition, measurement limitation, Operational Reality change, or unknown cause.

Attribution Must Not Be Invented

SIMULOS® establishes:

Observed deviation does not by itself establish the cause of deviation.

Where causation is not sufficiently supported, the status should remain:

Deviation Cause Undetermined.

This is preferable to false explanation.

Unresolved Deviation

A deviation may remain unresolved when: its cause is unknown, observations are incomplete, measurements conflict, reproduction fails, the real-world basis is uncertain, or multiple explanations remain plausible.

Unresolved deviations remain part of the governed correlation record.

No Average-Washing Principle

A critical SIMULOS® rule is:

Material Simulation-Reality Deviations must not disappear merely because aggregate averages remain acceptable.

A simulation could perform extremely well across 99% of conditions while failing in the 1% that matters most operationally.

No Selective Correlation Principle

Simulation-reality analysis should not preserve only those comparisons that demonstrate strong correspondence.

Material negative correlation evidence must remain visible.

Reality Change

A deviation may emerge because reality changed rather than because the simulation changed.

Therefore SDMM should preserve where possible:

Simulation Version

and:

Operational Reality State / Time Basis.

This allows later analysis to distinguish:

Simulation Drift

from:

Reality Drift

or:

Combined Drift.

Deviation Across Versions

Simulation versions may be compared against the same or equivalent reality basis.

For example:

Simulation v1 → Reality → Deviation Set A

Simulation v2 → Reality → Deviation Set B

This can reveal whether changes improved or worsened the correlation relationship.

Deviation Trend

Across repeated comparisons, deviations may:

Decrease

Remain Stable

Increase

Shift Location

Change Direction

or:

Become Uncertain.

Deviation trend can become an important governance signal for SRAM.

Deviation Record

A governed deviation record may contain:

Deviation ID

Simulation ID / Version

Reality Reference

Scenario / Condition

Correlation Criterion

Simulation Observation

Reality Observation

Deviation Measure

Deviation Direction

Deviation Magnitude

Deviation Classification

Materiality

Frequency

Affected Element

Known Dependencies

Probable Basis where supported

Uncertainty

Status

Required Follow-Up Relationship to SRCM

SRCM establishes:

what simulation observation and reality observation may legitimately be compared.

SDMM governs:

the material difference revealed by that comparison.

Without SRCM, a deviation may be based on incomparable observations.

Relationship to SCCM

SCCM establishes the correlation criteria and thresholds.

SDMM applies those criteria to structure and classify observed differences.

Therefore:

Correlation Criterion → Comparison → Deviation.

Relationship to SRAM

SDMM identifies where the simulation-reality relationship differs materially.

SRAM uses that governed deviation basis to determine where and how Reality Alignment should be examined.

Therefore:

Deviation Discovery precedes Alignment Action.

Relationship to SCUM

Some deviation measurements remain uncertain.

SCUM governs uncertainty in the broader simulation-to-reality correlation relationship.

SDMM must therefore preserve uncertainty rather than converting uncertain deviation into false precision.

Relationship to ORA™

ORA™ governs Operational Reality.

SDMM does not redefine reality.

It consumes sufficiently governed reality observations or descriptions and governs the deviation relationship between those observations and the simulation.

Relationship to VALIDOS®

SDMM does not determine whether the simulation is valid for a particular Validation Purpose.

It produces structured Simulation-Reality Deviation evidence that VALIDOS® may subsequently use.

Relationship to INTEGROS®

SDMM does not own general integrity governance.

INTEGROS® may govern integrity properties of the observations, measurements, records, or artifacts used in deviation analysis.

Autonomous Systems Application

For autonomous systems, SDMM can identify where simulated autonomous behavior diverges from operational behavior.

Material deviation regions may include: perception under degraded conditions, agent interaction, human response, recovery behavior, escalation, decision timing, unexpected environmental conditions, and long-horizon autonomous behavior.

This matters because a system can demonstrate excellent average simulation performance while diverging precisely in the operational conditions where autonomous governance is most important.

Space and Mission Simulation Application

Space systems often depend heavily on simulation because many mission conditions cannot be repeatedly reproduced in Operational Reality.

SDMM can govern differences revealed through: hardware tests, flight data, telemetry, prior missions, environmental measurements, subsystem tests, and operational mission observations.

A simulation may correlate strongly for nominal trajectory behavior while materially deviating in: thermal extremes, radiation response, communication degradation, sensor behavior, propulsion anomalies, subsystem interactions, or autonomous recovery.

The deviation record makes those limits explicit rather than allowing nominal correlation to represent the entire mission envelope.

Minimum Implementation Framework

1. Establish the Comparison Basis

Use governed simulation-reality comparisons from SRCM. 2. Apply Correlation Criteria

Determine which SCCM criteria govern the comparison. 3. Identify Deviations

Detect materially relevant differences. 4. Measure the Difference

Determine magnitude, direction, timing, distribution, or another applicable deviation measure. 5. Localize the Deviation

Identify the affected: condition, scenario, component, state, interaction, environment, or output.

6. Classify the Deviation

Determine whether it is: systematic, conditional, intermittent, localized, cumulative, or another relevant type.

7. Determine Materiality

Assess whether the deviation materially affects the intended simulation-reality relationship. 8. Identify Dependencies and Propagation

Determine whether other deviations or conditions contribute to or result from it. 9. Assess Attribution

Identify probable causes only where supported.

Otherwise preserve:

Cause Undetermined. 10. Preserve Uncertainty

Do not express deviation more precisely than the evidence permits. 11. Document the Deviation

Create a governed Simulation-Reality Deviation Record. 12. Transfer to Reality Alignment

Provide the structured deviation basis to SRAM.

Governance Outputs

SDMM may produce: Simulation-Reality Deviation ID, Simulation-Reality Deviation Record, Deviation Measurement Record, Deviation Magnitude Assessment, Deviation Direction Assessment, Deviation Localization Record, Temporal Deviation Record, Spatial Deviation Record, Behavioral Deviation Record, Environmental Deviation Record, Interaction Deviation Record, Trajectory Deviation Record, Systematic Deviation Record, Conditional Deviation Record, Intermittent Deviation Record, Deviation Frequency

Assessment, Deviation Materiality Assessment, Threshold Deviation Record, Deviation Cluster Record, Deviation Dependency Map, Deviation Propagation Map, Deviation Attribution Record, Unresolved Deviation Record, Reality Change Record, Cross-Version Deviation Comparison, Deviation Trend Record, and Governed Simulation-Reality Deviation Basis.

Use Case 1 — Autonomous Vehicle Simulation

An autonomous driving system performs strongly in simulation and demonstrates high aggregate correspondence with road testing.

SDMM reveals, however, that under the combined conditions of:

heavy rain + low illumination + partially obscured road markings

the simulated perception stack consistently identifies lane boundaries earlier than the real system.

The deviation is classified as:

Conditional + Systematic + Material.

Instead of concluding that the entire simulation is inaccurate, SDMM localizes the problem to a specific operational region and transfers the governed deviation to Reality Alignment. Use Case 2 — Spacecraft Autonomous Recovery

A spacecraft simulation predicts successful autonomous recovery from temporary communication loss across a broad mission scenario set.

Operational telemetry from comparable subsystem and mission conditions shows that real recovery begins materially later when:

low power state + communication degradation + thermal constraint

occur simultaneously.

SDMM identifies a conditional temporal and behavioral deviation that is largely invisible in nominal mission correlation statistics.

The deviation becomes a governed input to SRAM without prematurely declaring either the simulation or mission system invalid.

Architectural Position

SDMM is the third module of the Simulation-to-Reality Correlation Standard (SRCS).

The internal progression is:

SRCM — Simulation-Reality Comparison

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SCCM — Simulation Correlation Criteria

↓

SDMM — Simulation-Reality Deviation

↓

SRAM — Reality Alignment

↓

SCUM — Correlation Uncertainty

SDMM occupies the critical point where:

observed correspondence becomes structured knowledge about non-correspondence.

Foundational Principle

Simulation similarity is not reality equivalence, and aggregate correlation must never conceal the specific conditions under which simulation and reality materially diverge.

Canonical Closing Statement

Simulation-Reality Deviation Module (SDMM) governs how material differences between simulation and Operational Reality are identified, measured, localized, classified, contextualized, traced, and preserved. By distinguishing deviation from failure, global correlation from local divergence, magnitude from materiality, observation from attribution, and simulation change from reality change, SDMM prevents strong aggregate performance from concealing the operational regions where simulation correspondence weakens or fails. It transforms simulation-reality difference from an informal observation into a governed deviation basis capable of supporting Reality Alignment, correlation uncertainty governance, validation, audit, and future simulation improvement without falsely treating simulation similarity as reality equivalence.

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